

PROJECT AND TEAM INFORMATION

Project Title

CAMPUS RESCUE – Emergency Response & Resource Management Simulator

Student / Team Information

Team Name: Team ID:	CODE RUSHERS DSCPP-III-2026-T251
Team member 1 (Team Lead) Student ID: 2510370098	Gairola, Anirudh – 2510370098 2510370098@geu.ac.in 
Team member 2 Student ID: 2510011010	Sain, Anjali – 2510011010 2510011010@geu.ac.in 
Team member 3 Student ID: 2510012243	Mukherjee, Pratham – 2510012243 2510012243@geu.ac.in 
Team member 4 Student ID: 2510382260	Shai, Bishesh – 2510382260 2510382260@geu.ac.in 

Faculty: Ms. Nishtha Bhatt, Assistant Professor, CSE

Email: nishthabhatt.cse@geu.ac.in

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

College campuses may face medical emergencies, fires, accidents and security incidents. Effective response requires quick prioritization, suitable team assignment, efficient routing and proper resource tracking.

Campus Rescue provides a safe academic simulation of this process. It applies C++ Object-Oriented Programming to represent emergencies, teams, resources and locations as classes. Queues, priority queues, linked lists and stacks organize information, while graph algorithms help find efficient routes.

The project connects OOP, Data Structures and Algorithms with a practical problem while developing programming, teamwork, testing and problem-solving skills.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Universities generally use campus security teams, emergency communication systems, dispatch procedures and notification platforms to manage incidents. Professional systems may also use mobile applications, communication networks and external emergency services.

Campus Rescue simplifies these concepts into an educational simulation. It uses predefined locations, simulated emergencies, response teams and resources. The system demonstrates prioritization, team assignment, route calculation, resource management and incident records using C++, OOP, Data Structures and Algorithms.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

- Develop a C++ console-based emergency simulation.
- Apply classes, encapsulation, inheritance and polymorphism.
- Use Queue, Priority Queue, Linked List and Stack.
- Model campus locations using a weighted Graph.
- Implement BFS, DFS and Dijkstra's algorithm.
- Manage teams, resources and incident history.
- Provide searching, sorting, file storage and reports.
- **Milestones:** Research → Design → Core Coding → DSA → Routing → Integration → Testing → Documentation and Demo.

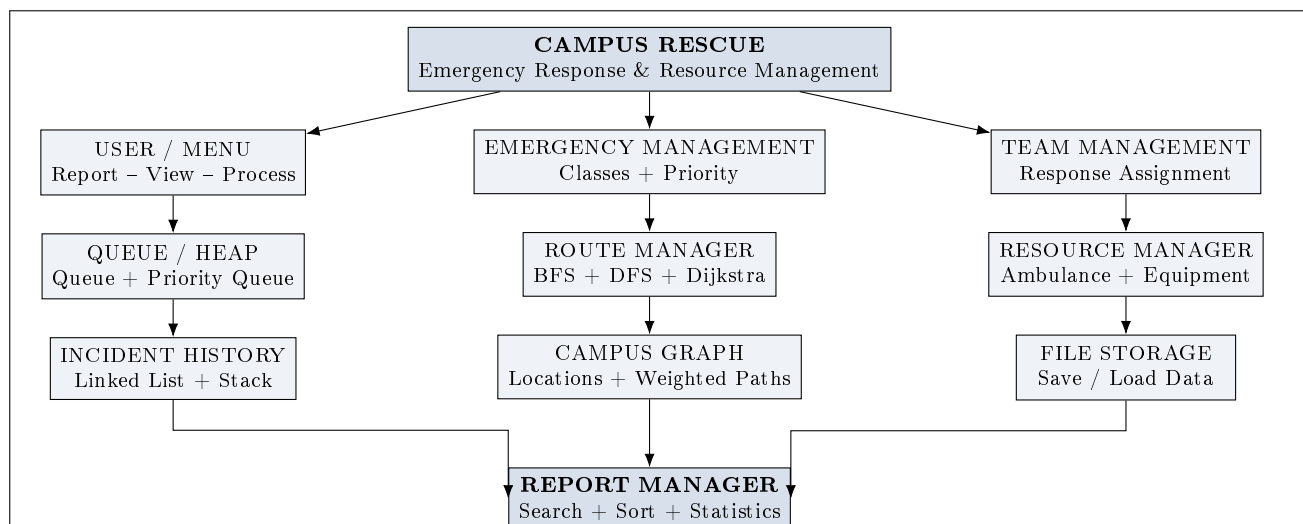
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

The system will be developed as a modular C++ console application. Emergencies, response teams, resources and locations will be represented using classes. Inheritance and polymorphism will represent specialized emergency and response-team types. A Queue will manage incoming cases, while a Priority Queue will process incidents according to severity. Linked Lists will store dynamic records and a Stack can support recent actions. The campus will be represented as a weighted Graph. BFS and DFS will provide traversal, while Dijkstra's algorithm will find shortest routes. File handling will store system data. The project will use C++, VS Code/Visual Studio and Git/GitHub. Modules will be tested independently before final integration and demonstration.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The project will deliver a functional C++ console-based Campus Rescue simulation.

- OOP classes for emergencies, teams, resources and locations.
- Queue, Priority Queue, Linked List and Stack applications.
- Campus Graph with BFS, DFS and Dijkstra.
- Emergency prioritization and team/resource management.
- Incident history, searching, sorting and file storage.
- Basic reports, test cases, documentation, PPT and final demo.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

- Campus locations and distances are predefined.
- Emergencies are manually entered simulated cases.
- Teams have predefined capabilities and availability.
- Resources are limited and predefined.
- The system is an academic simulation, not a real emergency-dispatch system.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. *cppreference – C++ Classes:*
<https://en.cppreference.com/cpp/language/classes.html>
2. *cppreference – C++ Inheritance:*
<https://en.cppreference.com/book/intro/inheritance.html>
3. *cppreference – C++ Polymorphism:*
<https://cppreference.com/book/intro/polymorphism.html>
4. *cppreference – std::priority_queue:*
https://en.cppreference.com/w/cpp/container/priority_queue/
5. *University of Iowa – Emergency Management:*
<https://safety.uiowa.edu/emergency-management>
6. *University of Iowa – Emergency Notifications:*
<https://safety.uiowa.edu/emergency-notifications>
7. *CrisisGo – Higher Education Safety:*
<https://www.crisisgo.com/higher-education>